

SCADA Hub

Feature Changelog · Sessions 1–25 · 2026-05-10 through 2026-05-16

OVERVIEW

Eight study guide sites (Modbus, OPC UA, DNP3, IEC 61131-3, PID Controllers, SEL RTAC, Ignition SCADA, Wireshark) rebuilt and relaunched as a production platform in one week. This document summarizes every feature shipped, infrastructure change made, and gap discovered across all 25 sessions.

PLATFORM INFRASTRUCTURE

1. **NEW** Vercel Deployment — All 9 Projects

All 8 study guides and the scada-hub landing page deployed to Vercel prod with custom `vercel.json` per project. Build command standardized: `GITHUB_PAGES= npm run build` (empty var sets Vite base to `/` for Vercel path compatibility). GitHub Pages continues to serve from the `docs/` folder for backward compatibility.

2. **FIX** GitHub Pages Branch Mismatch — DNP3

GitHub Pages was configured to serve from `main` while the DNP3 repo lived on `master`. All pushes were invisible. Fixed via `gh api PUT /repos/BarbosaRicardo/dnp3-study-guide/pages`. Added `.nojekyll` to all 8 guide `docs/` folders to prevent Jekyll from intercepting JS asset requests.

3. **FIX** scada-hub.vercel.app Blank Page

`AuthProvider` was imported in `App.jsx` but never used to wrap the component tree. `GuideGrid` calls `useAuth()` → `useContext(AuthContext)` returns `null` → destructuring crash → blank page. Fixed by wrapping `<HubContent />` in `<AuthProvider>`.

4. **INFRA** Supabase Row-Level Security — Quiz Submissions

Added `user_id` column to `quiz_submissions` table and enabled RLS. Policy: authenticated users read only their own rows; admin metadata bypass for full read access. Migration: `supabase/20260514_002_quiz_user_rls.sql`.

5. **INFRA** Cross-Ecosystem Links — All 8 Sidebars

Added an "All Courses" collapsible nav panel to every guide sidebar. Each guide shows all 8 sibling guides with icons and Vercel URLs. Links are environment-aware: `localhost` in dev, Vercel URLs in prod.

COURSE SYLLABI

6. **NEW** PDF Course Syllabi — All 8 Guides

LaTeX-compiled single-page syllabi added to all 8 guides. Each syllabus covers: course title, credit hours, prerequisites, required software, 10–12 topic sections in academic narrative format, and learning outcomes. Sidebar "Course Syllabus" button serves the PDF directly. Hosted at `site/public/syllabus.pdf` per guide.

7. **NEW** Syllabus Design Overhaul

Removed marketing footer text, badge graphics, and diagram boxes. Rewrote all topic sections in human prose (no AI-dash style). Added footer strip with guide accent color. Syllabus is now a professional academic document suitable for manager review and ADO submission.

UI / VISUAL FIXES

8. **FIX** Dark Theme Sweep — All 8 Guides

Replaced all light-mode Tailwind classes (`bg-white`, `bg-slate-50`, `bg-slate-100`, `text-navy-700`, `text-slate-700/800`, `border-slate-100/200`) with dark `rgba` inline styles across 67 files. Alternating table rows set to `rgba(255,255,255,0.05)`. Tap-hint dots in flashcard files corrected. Full rebuild and deploy for all 8 guides.

9. **FIX** DNP3 AnalogyCard — Dark Background

`AnalogyCard.jsx` used Tailwind gradient classes `from-purple-50 to-indigo-50` (white/light). Replaced with dark `rgba(139,92,246,0.06)` background and `rgba(139,92,246,0.25)` border. Button and icon wrapper also darkened.

10. **FIX** DNP3 QuizLevels — Emoji Removal + Dark Polish

Removed emoji fields from `LEVEL_META`. Replaced emoji spans with Lucide SVG icons (Lock, CheckCircle). Passed banner, resource drawer category badges, and empty state all converted to dark `rgba` styles. All `text-slate-700` → `text-slate-200`.

11. **FIX** Per-Guide Favicons

Each guide received a unique SVG favicon (colored to its accent — orange Modbus, cyan OPC UA, purple DNP3, etc.) replacing the generic Vite default.

12. **NEW** OG Image + Meta Tags — All 9 Projects

Open Graph `og:image`, `og:title`, `og:description`, and Twitter card meta tags added to all 9 `index.html` files. OG images generated

TRAINING FEATURE (NEW — SESSION 25)

13. **NEW** “More Training” Panel — All 8 Guides

TrainingPanel.jsx — a shared collapsible component — added to the landing/home page of every guide. Fetches live from Supabase: upcoming instructor-led and self-paced training events, and professional certification exams. Renders separate sections for Certifications (gold accent) and Upcoming Training (blue accent) with provider, format icon (online/in-person/hybrid/self-paced), and date range. Only events with `end_date ≥ today` (or self-paced, no date) are shown. Past events disappear automatically with no manual intervention.

14. **NEW** Vercel Cron Auto-Updater

api/refresh-training.js: serverless function running every Monday at 6 AM UTC via Vercel cron (configured in `vercel.json`). Each run: (1) queries Brave Search API for fresh training events per course, (2) routes results through Vercel AI Gateway (anthropic/claude-haiku-4.5) to parse into structured JSON, (3) upserts into Supabase `training_events` table keyed by URL — same event found again overwrites its row with updated dates, (4) HEAD-checks all stored URLs for broken links.

15. **NEW** Broken URL Email Alerts

When any stored training event URL returns a non-2xx status or times out, the cron sends an email alert to `ric3639@gmail.com` via the Resend API. Email lists each broken URL with course, provider, and HTTP status. No manual monitoring needed.

16. **INFRA** Supabase: `training_events` Table

Migration `20260516_003_training_events.sql`: table with RLS (public read, admin write), unique constraint on URL (enables upsert deduplication), and composite index on (course, end_date) for fast per-guide queries.

17. **NEW** 35 Verified 2026 Training Events Seeded

scripts/seed-training.js: initial dataset covering all 8 courses. Highlights: Ignition Core Certification (\$350), Ignition ICC 2026 (Sacramento, Sept 22–24), SANS ICS515 (Orlando, June 8–16 → feeds GICSP exam), SharkFest US 2026 (Nashville, July 18–23), ISA CCST and CAP certifications, Control Station PID workshops (June and September), DISTRIBUTECH DNP3 workshops (Feb 3), and SEL University RTAC courses (rolling enrollment).

COURSE CONTENT AUDIT (ALL 8 GUIDES)

18. **AUDIT** Lean Audit Pipeline — All 8 Guides

Deployed parallel Explore agents to audit every guide’s syllabus, web chapters, quiz data, and flashcard data. Output: counts summary + gap table only (no file dumps). Total gaps found: 30+ discrete content items across all 8 guides — each punchlisted as a numbered work item. Key finding: flashcard coverage averages 5–6 chapters per guide while quiz coverage averages 10, leaving 4–5 chapters per guide without flashcard reinforcement.

19. **AUDIT** Research Agent — Sources per Course

Background agent queried authoritative primary sources for all 8 courses: official specs, certification bodies, standard pricing, and publication access details. Results logged to `scada-hub/SOURCES.md`. Modbus spec suite fully documented (all free/downloadable). IEC 61158 access path clarified (Modbus appears under CPF 15 in IEC 61784-2, separate from the main IEC 61158 purchase).

20. **AUDIT** Watchdog Agent — Token Efficiency

Watchdog agent reviewed audit agent prompts and identified primary inefficiency: requesting verbatim card text vs card count added ~40% token overhead. Lean template (gap-table-only output, no file dumps) adopted for all remaining audits. Estimated token savings per audit run: ~40K tokens.

PUNCHLIST UPDATES

21. **INFRA** Punchlist: Removed Trimark Associates Reference

Login page branding item reworded to “Company A Brand Colors” — specific company name removed from all documentation.

22. **INFRA** Punchlist: R&D / AI Agent Skills Item Added

New Professional Development section added covering Claude AI (Claude Code, agent SDK, multi-agent orchestration), OpenAI Codex, and agentic toolchain evaluation. Deliverable: internal write-up of recommended adoption candidates.

23. **INFRA** Punchlist: 30+ Course Gap Items Added

Course Audit sections added for all 8 guides (Modbus, OPC UA, DNP3, IEC 61131-3, PID, RTAC, Ignition, Wireshark) with specific numbered items for each content gap. PDF recompiled to 5 pages after each update.